A Monte Carlo Study of Primordial Black Hole Flybys and the Earth–Mars Distance

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1 Overview

1.1 The question

A primordial black hole (PBH) of mass μ passes through the inner Solar System at about 200 km/s. It gives each planet a quick sideways push. The push is tiny, but it changes the planet's orbital period slightly, so over the following years the planet drifts ahead of or behind the place it would have been. The Earth–Mars distance is measured to about 0.1 m, so a large enough drift would show up as a slow wobble in that distance.

How massive does the black hole have to be for this wobble to be detectable, and how often would that happen if PBHs make up the dark matter?

1.2 Why a Monte Carlo

The size of the wobble depends on the direction the black hole comes from, how close it passes, and where Earth and Mars are in their orbits at that moment. We control none of these. A near-direct hit on Mars at the right orbital phase could be detectable even for a light black hole, and a distant flyby could be invisible even for a heavy one. So we draw many random flybys, compute the wobble for each, and count how many are detectable. This is a Monte Carlo calculation.

1.3 How the document is organized

Each part below follows the same pattern: it states how it builds on the previous part, shows the notebook code (the *chunk*), explains what the chunk does, and defines every function and variable in it.

Part	What it does	Builds on
1	One flyby, simple formula	Starting point
2	General physics functions	Part 1, with more complete physics
3	Check on one example flyby	Parts 1 and 2 agree
4	Random flybys	The functions of Part 2
5	Detection number q	The events of Part 4 and the functions of Part 2
6	Flyby rate	The q values of Part 5
7	Scan over mass and threshold	Parts 5 and 6
8	Windows and peaks	The scan of Part 7
9	Stability checks	Parts 4–8

Section 11 then gives the results and what they mean.

1.4 Units and notation

Lengths are in au, times in years, and masses of the black hole in the physics functions are in solar masses unless stated. In these units $GM_\odot = 4\pi^2$ au^3/yr^2, and 200 km/s $\approx$ 42.2 au/yr.

Symbol	Notebook name	Meaning
μ	`mu`, `mus`	Black hole mass (`mu` in kg, `mus` $= \mu/M_\odot$)
v_b	`vb`	Black hole speed, in au/yr
$\hat{\mathbf{v}}$	`vhat`	Unit vector along the black hole's direction of travel
$\mathbf{b}$	`bvec`	Impact vector: from the Sun to the closest point on the black hole's path ($\mathbf{b} \cdot \hat{\mathbf{v}} = 0$)
t_{ca}	`tca`	Time the black hole is closest to the Sun
a, Ω, ϕ	`mcPlanets`	A planet's orbit radius, angular speed and starting phase
$\mathbf{u}$	`u`	Velocity kick a planet receives
q	`mcQ`	Detection number
q_0	`q0`	Detection threshold

2 Part 1: One flyby with the simple formula

Where we start. This is the first section of the notebook. It handles a single flyby, with the black hole moving perpendicular to the plane of the orbits, and uses a simple formula for how the planets drift. Everything after this generalizes it.

2.1 Chunk 1a: the orbits of Earth and Mars

```
wearth = 2*Pi; phaseearth = (100.5*Pi)/180; aearth = 1.0;
xearth[t_] := aearth*Cos[wearth*t + phaseearth];
yearth[t_] := aearth*Sin[wearth*t + phaseearth];
amars = 1.5237; wmars = (2*Pi)/(amars^(3/2)); phasemars = (-4.57*Pi)/180;
xmars[t_] := amars*Cos[wmars*t + phasemars];
ymars[t_] := amars*Sin[wmars*t + phasemars];
txearth[t_] := -Sin[wearth*t + phaseearth];
tyearth[t_] := Cos[wearth*t + phaseearth];
txmars[t_] := -Sin[wmars*t + phasemars];
tymars[t_] := Cos[wmars*t + phasemars];
```

What it does. Puts Earth and Mars on circular orbits in the same plane around the Sun. A planet at radius a moves with angular speed from Kepler's law, $\Omega = 2\pi/a^{3/2}$ (in au and years), so

$$\mathbf{R}(t) = a\,(\cos(\Omega t + \phi),\ \sin(\Omega t + \phi)), \qquad \hat{\boldsymbol{\theta}}(t) = (-\sin(\Omega t + \phi),\ \cos(\Omega t + \phi)).$$

Here $\hat{\boldsymbol{\theta}}$ is the direction the planet is moving.

Name	Meaning
`wearth`, `wmars`	Angular speed Ω of each planet, in radians per year.
`aearth`, `amars`	Orbit radius in au.
`phaseearth`, `phasemars`	Position angle at $t = 0$, in radians (the values at that moment from the JPL table).
`xearth[t]`, `yearth[t]`, `xmars[t]`, `ymars[t]`	Position of each planet at time `t`.
`txearth[t]`, `tyearth[t]`, `txmars[t]`, `tymars[t]`	Components of the unit vector $\hat{\boldsymbol{\theta}}$ along each planet's direction of motion.

2.2 Chunk 1b: the black hole and the impulse

```
vb = (200000*365.25*24*60*60)/(1.496*(10^11));
alpha = 0; beta = 0; tzero = 0;
vbx = vb*Sin[alpha]*Cos[beta];
vby = vb*Sin[alpha]*Sin[beta]; vbz = vb*Cos[alpha];
xzero = 1.5; yzero = -0.3;
mu = 10^18; msun = 2*(10^30);

rminsquareearth = ((xearth[tzero] - xzero)^2) + ((yearth[tzero] - yzero)^2) -
  (((xearth[tzero] - xzero)*(vbx/vb) + (yearth[tzero] - yzero)*(vby/vb))^2);
numfirstearth = xzero*txearth[tzero] + yzero*tyearth[tzero];
numsecondearth = ((vbx/vb)*txearth[tzero] + (vby/vb)*tyearth[tzero])*
  ((xearth[tzero] - xzero)*(vbx/vb) + (yearth[tzero] - yzero)*(vby/vb));
numeearth = numfirstearth + numsecondearth;
prefactearth = (4*Pi*(mu/msun))/vb;
impulseearth = (prefactearth*numeearth)/rminsquareearth
```

The same lines with `earth` replaced by `mars` (and `xearth` etc. by `xmars` etc.) give the impulse on Mars.

What it does. Describes the black hole and computes the tangential impulse each planet receives. The black hole passes through the orbital plane at (x_0, y_0) moving with velocity $v_b\,(\sin\alpha\cos\beta,\ \sin\alpha\sin\beta,\ \cos\alpha)$. With $\alpha = 0$ it moves straight along z, perpendicular to the planets' plane.

Only the push *along* a planet's orbit gives the steady drift described in the overview, so the code computes just that part, I_θ/m (the along-orbit impulse per unit planet mass). The planet is treated as standing still during the flyby, at its position $\mathbf{R}$ at time `tzero`. With $\mathbf{r}_0 = (x_0, y_0, 0)$ the point where the black hole crosses the plane, and $\hat{\mathbf{T}}$ the planet's direction of motion,

$$R_{\min}^2 = |\mathbf{R} - \mathbf{r}_0|^2 - \big((\mathbf{R} - \mathbf{r}_0)\cdot\hat{\mathbf{v}}\big)^2, \qquad \frac{I_\theta}{m} = \text{prefactor} \times \frac{\mathbf{r}_0\cdot\hat{\mathbf{T}} + (\hat{\mathbf{v}}\cdot\hat{\mathbf{T}})\big((\mathbf{R} - \mathbf{r}_0)\cdot\hat{\mathbf{v}}\big)}{R_{\min}^2}.$$

$R_{\min}$ is the closest distance between the planet and the black hole's path. The impulse falls off as $1/R_{\min}$, since the numerator has size $R_{\min}$ in the typical case where the black hole path is perpendicular to the line to the planet.

Name	Meaning
`vb`	Black hole speed, 200 km/s converted to au/yr.
`alpha`, `beta`	Colatitude and longitude of the velocity direction, in radians.
`tzero`	Time of the flyby, years.
`vbx`, `vby`, `vbz`	Components of the black hole velocity.
`xzero`, `yzero`	Where the black hole crosses the orbital plane, in au.
`mu`, `msun`	Black hole mass and the (rounded) mass of the Sun, in kg.
`rminsquareearth`	$R_{\min}^2$ for Earth: square of the closest distance between Earth and the black hole's path.
`numfirstearth`, `numsecondearth`	The two terms in the numerator of the impulse formula.
`numeearth`	Their sum, the full numerator.
`prefactearth`	The constant in front, proportional to $(\mu/M_\odot)/v_b$.
`impulseearth`	I_θ/m for Earth, in au/yr.

2.3 Chunk 1c: how the planets drift and the Earth–Mars distance

```
deltaxearth[t_] := -3*impulseearth*(t - tzero)*txearth[t]*HeavisideTheta[t - tzero];
deltayearth[t_] := -3*impulseearth*(t - tzero)*tyearth[t]*HeavisideTheta[t - tzero];
(* same two lines for Mars *)

distearthmars[t_] := Sqrt[(xearth[t] - xmars[t])^2 + (yearth[t] - ymars[t])^2];
nearthmarsx[t_] := (xearth[t] - xmars[t])/distearthmars[t];
nearthmarsy[t_] := (yearth[t] - ymars[t])/distearthmars[t];
autom = 1.496*(10^11);
deltaearthmars[t_] := autom*(((deltaxearth[t] - deltaxmars[t])*nearthmarsx[t]) +
   ((deltayearth[t] - deltaymars[t])*nearthmarsy[t]));
tranestimate[t_] := autom*prefactmars*(t - tzero)*HeavisideTheta[t - tzero]/Sqrt[rminsquaremars];
```

What it does. A push along the orbit raises or lowers the orbit and so changes the orbital period (a higher orbit is slower). The planet then drifts steadily ahead of or behind where it would have been, and the drift grows linearly with time:

$$\Delta\mathbf{r}(t) = -3\,\frac{I_\theta}{m}\,(t - t_0)\,\hat{\boldsymbol{\theta}}(t), \qquad t > t_0.$$

Only the part of the relative drift that lies along the line joining Earth and Mars changes their distance (to first order):

$$\Delta r_{12}(t) = \hat{\mathbf{n}}_{12} \cdot (\Delta \mathbf{r}_E - \Delta \mathbf{r}_M), \qquad \hat{\mathbf{n}}_{12} = \frac{\mathbf{r}_E - \mathbf{r}_M}{|\mathbf{r}_E - \mathbf{r}_M|}.$$

The result is converted from au to metres with `autom`. For comparison, `tranestimate` is the simple estimate $\delta v\, \Delta t$ used by Tran et al., where the speed change is the impulse at the closest approach, divided by the planet's mass.

Name	Meaning
`deltaxearth[t]`, `deltayearth[t]`	x and y drift of Earth at time `t`. The `HeavisideTheta` factor makes it zero before the flyby.
`deltaxmars[t]`, `deltaymars[t]`	Same for Mars.
`distearthmars[t]`	Unperturbed Earth–Mars distance.
`nearthmarsx[t]`, `nearthmarsy[t]`	Components of the unit vector $\hat{\mathbf{n}}_{12}$ from Mars to Earth.
`autom`	Metres per au.
`deltaearthmars[t]`	Change in the Earth–Mars distance, in metres.
`tranestimate[t]`	The Tran et al. style estimate of the same quantity, for comparison.

What Part 1 cannot do. It treats a single flyby with a single simple geometry. It ignores how the Sun itself is pushed, it assumes the planets are fixed during the flyby, and it keeps only the growing drift. Part 2 removes these limits.

3 Part 2: General physics functions

Building on Part 1. Part 1 gave the idea: a push along the orbit makes the planet drift, and the drift changes the Earth–Mars distance. To run many random flybys we need the same physics written as functions that take any direction and any impact vector, and that include more of the physics. This part builds those functions step by step.

3.1 Chunk 2a: setup

```
msun = 1.98847*^30;
gmsun = 4 Pi^2;
mcSunKick = True;
mcFloor = 10.^-8;
mcPlanets = <|"Earth" -> {aearth, wearth, phaseearth},
             "Mars" -> {amars, wmars, phasemars}|>;
mcXY[name_, t_] := Module[{a, w, ph}, {a, w, ph} = mcPlanets[name];
   {a Cos[w t + ph], a Sin[w t + ph]}];
```

What it does. Sets the constants and stores each planet's orbit in one table, reusing the values from Part 1, so that later functions can look a planet up by name.

Name	Meaning
`msun`	Mass of the Sun in kg (overrides the rounded value of Part 1). Used to convert a black hole mass in kg to solar masses.
`gmsun`	$GM_\odot$ in au^3/yr^2, equal to $4\pi^2$.
`mcSunKick`	Switch. If `True`, the Sun's own kick is subtracted from each planet's kick (see Chunk 2b).
`mcFloor`	Smallest allowed value of $\lvert\mathbf{d}\rvert^2$ in the kick formula, in au^2. It stops a flyby that lands almost on a planet from giving an infinite kick.
`mcPlanets`	Table of each planet's orbit: radius a, angular speed Ω and phase.
`mcXY[name, t]`	Position (x, y) of the named planet at time `t`.

3.2 Chunk 2b: the kick from the black hole

```
mcKick[R_, bvec_, vhat_, mus_] := Module[{d},
   d = bvec - (R - (R.vhat) vhat);
   (2 gmsun mus/vb) d/Max[d.d, mcFloor]];
```

What it does. Gives the velocity kick that a passing black hole delivers to a body at position $\mathbf{R}$. The black hole moves in a straight line much faster than any planet, so the planet is treated as standing still during the flyby and other forces are ignored for that short time (the impulse approximation). For a body at perpendicular distance b from the path, the sideways force is $G\mu m\, b/(b^2+v_b^2t^2)^{3/2}$. Integrating over the whole flyby gives a total momentum of $2G\mu m/(v_b b)$, so the velocity kick is $2G\mu/(v_b b)$, directed toward the path. The planet's mass cancels. In vector form,

$$\mathbf{u} = \frac{2G\mu}{v_b}\,\frac{\mathbf{d}}{|\mathbf{d}|^2}, \qquad \mathbf{d} = \mathbf{b} - \big(\mathbf{R} - (\mathbf{R}\cdot\hat{\mathbf{v}})\hat{\mathbf{v}}\big).$$

The kick is stronger the closer the pass, and it is proportional to the black hole mass μ. That second fact is used later: every displacement and every detection number is proportional to μ, so the Monte Carlo is run once at a reference mass and rescaled.

The Sun's kick. The black hole pulls the Sun as well. If the flyby is far away, Earth, Mars and the Sun all receive almost the same kick, and moving together does not change the orbits. What changes the orbits is the *difference* between a planet's kick and the Sun's, which is why the next function subtracts the Sun's kick (found by calling `mcKick` at $\mathbf{R} = \mathbf{0}$). This is new compared with Part 1, and it matters a great deal: without it a far flyby looks like a large push on both planets, and the typical detection number comes out about 40 times too large.

Name	Meaning
`R`	Position of the body (a 3-vector). The Sun is at the origin.
`bvec`	Impact vector: from the Sun to the closest point on the black hole's path, perpendicular to `vhat`.
`vhat`	Unit vector along the black hole's direction of travel.
`mus`	Black hole mass in solar masses.
`d`	Vector from the body to the closest point of the path, perpendicular to `vhat`.
`gmsun`, `vb`, `mcFloor`	As defined above.

3.3 Chunk 2c: how a planet responds to the kick

```
mcDisp[name_, {vhat_, bvec_, tca_}, mus_, tg_] :=
 Module[{a, w, ph, ti, R, u, rh, th, ur, ut, tau, xi, eta, ang},
  {a, w, ph} = mcPlanets[name];
  ti = tca;
  Do[R = a {Cos[w ti + ph], Sin[w ti + ph], 0.};
     ti = tca + (R.vhat)/vb, {2}];
  R = a {Cos[w ti + ph], Sin[w ti + ph], 0.};
  u = mcKick[R, bvec, vhat, mus] -
      If[mcSunKick, mcKick[{0., 0., 0.}, bvec, vhat, mus], 0.];
  rh = {Cos[w ti + ph], Sin[w ti + ph], 0.};
  th = {-Sin[w ti + ph], Cos[w ti + ph], 0.};
  ur = u.rh; ut = u.th;
  tau = UnitStep[tg - ti] (tg - ti);
  xi = (ur/w) Sin[w tau] + (2 ut/w) (1 - Cos[w tau]);
  eta = (2 ur/w) (Cos[w tau] - 1) + (4 ut/w) Sin[w tau] - 3 ut tau;
  ang = w tg + ph;
  {xi Cos[ang] - eta Sin[ang], xi Sin[ang] + eta Cos[ang]}];
```

What it does. Follows one planet after the kick and returns how far it has moved from its unperturbed orbit at each time in the list `tg`.

1. *When is the kick felt?* The black hole is at $\mathbf{b} + \hat{\mathbf{v}}\, v_b(t - t_{ca})$, so it is closest to a planet at $\mathbf{R}$ at time $t_i = t_{ca} + \mathbf{R} \cdot \hat{\mathbf{v}}/v_b$. The planet position depends on t_i, so this is repeated twice. (Part 1 used one common time for both planets.)
2. *What is the kick?* Compute it, subtract the Sun's kick, and split the result into a radial part u_r and an along-orbit part u_θ.
3. *How does the planet respond?* For small displacements from a circular orbit, with $\tau = t - t_i \geq 0$ (zero before the kick), ξ the displacement outward and η the displacement ahead along the orbit, in a frame that turns with the planet:

$$\xi = \tfrac{u_r}{\Omega} \sin \Omega\tau + \tfrac{2u_\theta}{\Omega}(1 - \cos \Omega\tau),$$
$$\eta = \tfrac{2u_r}{\Omega}(\cos \Omega\tau - 1) + \tfrac{4u_\theta}{\Omega} \sin \Omega\tau - 3u_\theta\tau.$$

   These come from the epicycle equations in the impulse response notes.
4. Rotate (ξ, η) back to the fixed x, y axes.

The term $-3u_\theta\tau$ is the same steady drift as in Part 1 (with $u_\theta = -I_\theta/m$ in the sign convention of the notes). The other terms only oscillate at the orbital frequency with a size of order u/Ω. They were dropped in Part 1 and are kept here.

Name	Meaning
`name`	`"Earth"` or `"Mars"`.
`tca`	Time the black hole is closest to the Sun.
`tg`	List of times at which the displacement is wanted.
`a, w, ph`	The planet's orbit radius, angular speed Ω and phase.
`ti`	Time the black hole is closest to this planet, when the planet gets its kick.
`u`	Kick on the planet, with the Sun's kick subtracted if `mcSunKick` is `True`.
`rh, th`	Unit vectors along the planet's radius and along its direction of motion, at time `ti`.
`ur, ut`	Radial and along-orbit parts of the kick.
`tau`	Time since the kick, set to zero before it.
`xi, eta`	Displacement outward and ahead along the orbit, in the turning frame.
`ang`	Angle of the planet's unperturbed position, used to rotate back to x, y.

A simple version, for checking. The function `mcDispSecular` keeps only the steady drift, as in Part 1, but in the general event format and with the planet position taken at t_{ca}:

```
mcDispSecular[name_, {vhat_, bvec_, tca_}, mus_, tg_] :=
 Module[{a, w, ph, R, u, th0, ut, tau},
  {a, w, ph} = mcPlanets[name];
  R = a {Cos[w tca + ph], Sin[w tca + ph], 0.};
  u = mcKick[R, bvec, vhat, mus] -
      If[mcSunKick, mcKick[{0., 0., 0.}, bvec, vhat, mus], 0.];
  th0 = {-Sin[w tca + ph], Cos[w tca + ph], 0.};
  ut = u.th0; tau = UnitStep[tg - tca] (tg - tca);
  {3 ut tau Sin[w tg + ph], -3 ut tau Cos[w tg + ph]}];
```

It returns $\Delta\mathbf{r} = -3u_\theta(t - t_{ca})\hat{\boldsymbol{\theta}}(t)$ and is used in Part 3 to check `mcDisp`.

3.4 Chunk 2d: the Earth–Mars distance

```
mcDeltaR[partner_, ev_, mus_, tg_, disp_ : mcDisp] :=
 Module[{dE, dX, xe, ye, xp, yp, d},
  dE = disp["Earth", ev, mus, tg]; dX = disp[partner, ev, mus, tg];
  {xe, ye} = mcXY["Earth", tg]; {xp, yp} = mcXY[partner, tg];
  d = Sqrt[(xe - xp)^2 + (ye - yp)^2];
  autom ((xe - xp) (dE[[1]] - dX[[1]]) + (ye - yp) (dE[[2]] - dX[[2]]))/d];
```

What it does. The same projection as in Part 1, $\Delta r_{12} = \hat{\mathbf{n}}_{12}\cdot(\Delta\mathbf{r}_E - \Delta\mathbf{r}_M)$, now written as one function that works with any flyby and with either displacement model. This is the main output of the physics: a list of Earth–Mars distance changes, in metres, at the times in `tg`.

Name	Meaning
`partner`	The planet whose distance from Earth is measured (here `"Mars"`).
`ev`	One flyby, written `{vhat, bvec, tca}`.
`disp`	Which displacement model to use. The default is `mcDisp`; `mcDispSecular` can be passed for checking.
`dE, dX`	Displacements of Earth and of the partner, each as x and y lists over `tg`.
`xe, ye, xp, yp`	Unperturbed positions of Earth and the partner.
`d`	Unperturbed Earth–partner distance.
`autom`	Metres per au, as in Part 1.

4 Part 3: Check on one example flyby

Building on Parts 1 and 2. Before generating random flybys, we make sure the general functions of Part 2 agree with the simple calculation of Part 1 on the same flyby.

```
mcEvNotebook = Module[{vh, P0, bv, tc},
   vh = N[{vbx, vby, vbz}/vb];
   P0 = N[{xzero, yzero, 0}];
   bv = P0 - (P0.vh) vh;
   tc = tzero - (P0.vh)/vb;
   {vh, bv, tc}];
tgB = Range[tzero - 2 + 0.0013, tzero + 10, 10/365.25];
mus0 = mu/msun;
dMars = mcDeltaR["Mars", mcEvNotebook, mus0, tgB];

mcSunKick = False;
dFull = mcDeltaR["Mars", mcEvNotebook, mus0, tgB];
dSec = mcDeltaR["Mars", mcEvNotebook, mus0, tgB, mcDispSecular];
mcSunKick = True;
{Max[Abs[Pick[dFull, Thread[tgB < tzero]]]], Max[Abs[dSec]],
 Max[Abs[dFull - dSec]]}
```

What it does. Part 1 describes a flyby by where the black hole crosses the ecliptic and its velocity. Part 2 needs the direction, the impact vector and the time of closest approach to the Sun, so the first lines convert one description into the other. The flyby is then computed in two ways, the full response and the simple drift formula, with the Sun's kick switched off so that both use the same physics. The three printed numbers are the largest full value before the flyby (it must be exactly zero), the largest simple-formula value, and the largest difference between the two. They should agree to roughly 10 percent at late times, because the simple formula drops the oscillating terms.

Name	Meaning
`vh`	Unit vector along the black hole velocity, from `vbx`, `vby`, `vbz`.
`P0`	The point where the black hole crosses the ecliptic, as a 3-vector.
`bv`	Impact vector: `P0` with its component along the direction of travel removed.
`tc`	Time of closest approach to the Sun.
`mcEvNotebook`	The example flyby in the form `{vhat, bvec, tca}`.
`tgB`	Time grid: from 2 yr before to 10 yr after `tzero`, every 10 days.
`mus0`	Black hole mass of the example, in solar masses.
`dMars`	Earth–Mars distance change for the example, in metres.
`dFull`, `dSec`	Same flyby with the full response and with the simple formula (Sun's kick off).

5 Part 4: Random flybys

Building on Part 2. The functions of Part 2 turn *one* flyby, `{vhat, bvec, tca}`, into a distance change. A real black hole would arrive from a random direction at a random time, so we now generate many random events in that same format.

```
mcSample[bmax_, t0_, Tobs_] := Module[{c, ph, v, h, e1, e2, b, psi, bv},
  c = RandomReal[{-1, 1}]; ph = RandomReal[{0, 2 Pi}];
  v = {Sqrt[1 - c^2] Cos[ph], Sqrt[1 - c^2] Sin[ph], c};
  h = If[Abs[c] < 0.9, {0., 0., 1.}, {1., 0., 0.}];
  e1 = Normalize[Cross[v, h]]; e2 = Cross[v, e1];
  b = bmax Sqrt[RandomReal[]]; psi = RandomReal[{0, 2 Pi}];
  bv = b (Cos[psi] e1 + Sin[psi] e2);
  {v, bv, RandomReal[{t0, t0 + Tobs}]}];
mcBmax = 50.; mcT0 = 0.; mcTobs = 26.; mcN = 200000;
SeedRandom[1];
mcEvents = Table[mcSample[mcBmax, mcT0, mcTobs], {mcN}];
```

What it does. Each event is made of:

- a **direction** $\hat{\mathbf{v}}$, uniform over the sphere (the cosine of the polar angle uniform in $[-1, 1]$, the azimuth uniform in $[0, 2\pi]$);
- an **impact vector b**, uniform over a disk of radius $b_{\max} = 50\,\text{au}$ perpendicular to $\hat{\mathbf{v}}$. The distance is $b_{\max}\sqrt{\text{uniform}[0,1]}$; the square root keeps the points evenly spread over the disk instead of crowded near the centre;
- a **closest-approach time** t_{ca}, uniform over the observing window.

The 50 au disk is the "target" that a black hole must pass through; flybys farther out have a negligible effect.

Name	Meaning
`c, ph`	Cosine of the polar angle and the azimuth of the direction, both drawn uniformly.
`v`	The direction (a unit vector), which becomes `vhat`.
`h`	Any vector not parallel to the direction, used to build two perpendicular directions.
`e1, e2`	Two unit vectors perpendicular to `v`; the impact vector is built from them.
`b, psi`	Distance and angle of the impact point within the disk.
`mcBmax`	Radius of the target disk, 50 au.
`mcT0, mcTobs`	Start and length of the observing window (26 yr).
`mcN`	Number of random events.
`mcEvents`	The list of all random events.

A check line then prints the number of events, the largest $|\mathbf{b}|$ (just under 50), the average z component of the direction (near 0 if directions are balanced), and the largest dot product of the impact vector with the direction (essentially 0, since they are perpendicular).

6 Part 5: The detection number q

Building on Parts 2 and 4. Part 2 gives the Earth–Mars distance change for one flyby, and Part 4 gives a list of flybys. We now reduce each flyby to one number that says whether it could be seen.

```
mcTg = Range[mcT0, mcT0 + mcTobs, 40/365.25];
mcTargets = <|"Mars" -> 0.1|>;
mcMuBase = 10.^18;
mcQ[ev_, mus_] := Sqrt[Max[Total[
   (mcDeltaR[#, ev, mus, mcTg]/mcTargets[#])^2 & /@ Keys[mcTargets]]]];
mcQfor[events_, mus_] := Table[mcQ[ev, mus], {ev, events}];
mcQ1 = mcQfor[mcEvents, mcMuBase/msun];
```

What it does. Imagine measuring the Earth–Mars distance every 40 days for 26 years with a ruler accurate to $\sigma_r = 0.1\,\mathrm{m}$. A flyby makes the distance wobble. For each event, the detection number is the largest wobble divided by the ruler's accuracy:

$$q = \max_t \frac{|\Delta r_{12}(t)|}{\sigma_r}.$$

If $q > 1$ the wobble is bigger than the measurement error and the flyby would be seen directly. If $q \ll 1$ it could be recovered only by careful signal processing, as in searches for gravitational waves, where signals far below the noise are extracted.

q is computed at one reference mass, $\mu = 10^{18}$ kg (10^{21} g). Because Δr_{12} is proportional to μ (Chunk 2b), the value at any other mass M is

$$q(M) = q_{\mathrm{base}}\,\frac{M}{M_{\mathrm{base}}}.$$

This is why one run is enough for all masses. This is the slow step of the notebook: 2×10^5 events took about 68 seconds.

Name	Meaning
`mcTg`	The measurement dates: every 40 days over the 26 years.
`mcTargets`	Measurement accuracy σ_r of each target planet, in metres. Only Mars is used.
`mcMuBase`	Reference black hole mass, in kg.
`mcQ[ev, mus]`	q for one event at mass `mus`.
`mcQfor[events, mus]`	q for a list of events.
`mcQ1`	List of q for every event, at the reference mass.

7 Part 6: Flyby rate and detectable events

Building on Part 5. The list `mcQ1` tells us which *fraction* of flybys are detectable at each mass. To turn that into a number of events we also need to know how often flybys happen.

```
mcGeVkg = 1.78266*^-27;
mcAuCm = autom*100.;
mcRhoDM = 0.4 mcGeVkg mcAuCm^3;
mcSigma = Pi mcBmax^2;
mcFlybyRate[Mkg_] := (mcRhoDM/Mkg) mcSigma vb;
mcCount[q_, Mkg_, q0_] := Total[UnitStep[q - q0 mcMuBase/Mkg]];
```

What it does. If all the dark matter is black holes of mass M, their number density is ρ_{DM}/M, and the rate at which they cross the target disk is

$$\Gamma(M) = \frac{\rho_{DM}}{M}\,\pi b_{\max}^2\, v_b.$$

The rate falls as $1/M$: heavier black holes are rarer. Given a threshold q_0 (how small a wobble counts as detected), an event is detectable at mass M when $q_{\text{base}} > q_0\, M_{\text{base}}/M$, which follows from $q(M) > q_0$. The function `mcCount` counts such events.

Name	Meaning
`mcGeVkg`	Kilograms per GeV/c^2.
`mcAuCm`	Centimetres per au.
`mcRhoDM`	Local dark matter density, 0.4 GeV/cm^3, converted to kg/au^3.
`mcSigma`	Area of the target disk in au^2.
`mcFlybyRate[Mkg]`	Number of black holes of mass `Mkg` crossing the disk per year.
`mcCount[q, Mkg, q0]`	Number of events in the list `q` whose q would exceed `q0` at mass `Mkg`.

As a check, the notebook prints the number of black holes inside 5.2 au at 10^{18} g, the flyby rate at 10^{21} g, and the number of flybys in 26 years.

8 Part 7: Scan over mass and threshold

Building on Parts 5 and 6. We now combine the detectable fraction (Part 5) with the flyby rate (Part 6), at every mass of interest and for several thresholds.

```
mcQ0List = {1, 10^-2, 10^-3, 10^-4};
mcMg = 10^Range[12., 25., 0.1];
mcMinValidG = 10.^18;
mcMinCount = 10;
mcRowsFor[q_, q0_] := Table[
  Module[{Mkg = Mg/1000., n = Length[q], c, p, G},
   c = mcCount[q, Mkg, q0]; p = c/n; G = mcFlybyRate[Mkg];
   {Mg, p, Sqrt[p (1 - p)/n], c, G p, G p mcTobs, Mg >= mcMinValidG}],
  {Mg, mcMg}];
mcScan = AssociationMap[mcRowsFor[mcQ1, #] &, mcQ0List];
```

What it does. For each mass from 10^{12} to 10^{25} g and each threshold, it builds one row. With c detectable events out of n, the detectable fraction is $p = c/n$, the detectable rate is Γp, and the expected number of detections in the window is $N_{\mathrm{det}} = \Gamma\, p\, T_{\mathrm{obs}}$. The statistical error on p is $\sqrt{p(1-p)/n}$.

The threshold $q_0 = 1$ is the conservative case, where the wobble must exceed the measurement error. The smaller values, 10^{-2} to 10^{-4}, are the benchmark range of Tran et al., which assume that signal processing can extract much weaker signals.

Validity. Below about 10^{18} g the black holes are so numerous that many flybys happen during the observing window and their signals overlap, so this method no longer applies (Tran et al.). Masses below `mcMinValidG` are marked not valid.

Name	Meaning
`mcQ0List`	The detection thresholds q_0.
`mcMg`	Grid of black hole masses, in grams, from 10^{12} to 10^{25} in steps of 0.1 dex.
`mcMinValidG`	Lowest mass where the method is valid (10^{18} g).
`mcMinCount`	Minimum number of events needed behind a window edge for it to be trusted.
`mcRowsFor[q, q0]`	One row per mass: mass in g; p; the error on p; the number of events c behind p; the detectable rate per year; the expected detections in the window; and whether the mass is valid.
`mcScan`	The rows for every threshold.

9 Part 8: Windows and peaks

Building on Part 7. The scan is a long table. This part reduces it to the quantities we report.

```
mcWindow[rows_] := With[{ok = Select[rows,
    #[[6]] >= 1 && #[[7]] && #[[4]] >= mcMinCount &]},
   If[ok === {}, None, {Min[ok[[All, 1]]], Max[ok[[All, 1]]]}]];
mcPeak[rows_] := First[MaximalBy[rows, #[[5]] &]][[1]];
Grid[Prepend[Table[{q0, mcWindow[mcScan[q0]], mcPeak[mcScan[q0]]},
   {q0, mcQ0List}], {"q0", "window for N_det >= 1 [g]",
   "peak-rate mass [g]"}], Frame -> All]
mcMmaxG = mcRhoDM mcSigma vb mcTobs*1000.
```

What it does. The *window* is the range of masses where at least one detection is expected in the 26 years, counting only valid masses and only edges backed by at least `mcMinCount` events, so that one unusual

event cannot set an edge. The *peak* is the mass where the detectable rate is highest. The notebook also plots the detectable rate against mass for each threshold (valid masses only), with a horizontal line at one detection over the window.

Name	Meaning
`mcWindow[rows]`	Smallest and largest valid mass with at least one expected detection, or `None` if there is no such mass.
`mcPeak[rows]`	Mass at which the detectable rate is largest.
`mcMmaxG`	Mass in g at which exactly one flyby is expected through the disk in the window, from $\rho_{DM}\,\pi b_{\max}^2\,v_b\,T_{\rm obs}$.

10 Part 9: Stability checks

Building on Parts 4–8. The results should not depend on arbitrary choices. This part tests two: the floor on $|\mathbf{d}|^2$ in the kick formula, and the particular random events drawn.

```
mcSubset[seed_] := (SeedRandom[seed];
   Table[mcSample[mcBmax, mcT0, mcTobs], {20000}]);
mcCheck[f_, events_] := Module[{old = mcFloor, q},
   mcFloor = f; q = mcQfor[events, mcMuBase/msun]; mcFloor = old;
   {f, Median[q], Max[q], mcWindow[mcRowsFor[q, 10^-4]],
    mcWindow[mcRowsFor[q, 1]]}];
evA = mcSubset[11];
Grid[Prepend[mcCheck[#, evA] & /@ {10.^-6, 10.^-8, 10.^-10},
  {"floor", "median q", "max q", "window q0=1e-4 [g]",
   "window q0=1 [g]"}], Frame -> All]
Grid[Prepend[{mcCheck[10.^-8, mcSubset[12]], mcCheck[10.^-8, mcSubset[13]]},
  {"floor", "median q", "max q", "window q0=1e-4 [g]",
   "window q0=1 [g]"}], Frame -> All]
```

What it does. It recomputes q for a fresh batch of 20,000 events, first with the floor changed by factors of 100 on the same events, then with the usual floor on batches from different seeds, and compares the median q, the largest q and the windows for $q_0 = 10^{-4}$ and $q_0 = 1$. If the windows stay the same, the floor and the particular random draw do not affect the answer. The largest q is allowed to change, because it is set by a single unusually close flyby.

Name	Meaning
`mcSubset[seed]`	A fresh batch of 20,000 random events from the given seed.
`mcCheck[f, events]`	Sets the floor to `f`, recomputes q for the events, restores the floor, and returns the floor, the median and largest q, and the windows for $q_0 = 10^{-4}$ and $q_0 = 1$.

11 Results and what they mean

The run used 2×10^5 random flybys, a black hole speed of 200 km/s, a 50 au target disk, a 26 year window, Mars as the target with $\sigma_r = 0.1$ m, and measurements every 40 days.

11.1 Results

Quantity	Value
Median q at 10^{21} g	0.0138
Largest q at 10^{21} g	591.5
Black holes inside 5.2 au at 10^{18} g	1.41
Flyby rate through the disk at 10^{21} g	0.79 per year (20.6 in 26 yr)
Mass with one expected flyby in 26 yr	2.06×10^{22} g
Window, $q_0 = 1$	None
Window, $q_0 = 10^{-2}$	10^{18} to 1.58×10^{22} g
Window, $q_0 = 10^{-3}$	10^{18} to 2.0×10^{22} g
Window, $q_0 = 10^{-4}$	10^{18} to 2.0×10^{22} g
Peak-rate mass, $q_0 = 1$, 10^{-2}, 10^{-3}, 10^{-4}	6.3×10^{19}, 6.3×10^{17}, 6.3×10^{16}, 6.3×10^{15} g

Stability checks (Part 9).

Test	Median q	Largest q	Windows
Floor 10^{-6}, 10^{-8}, 10^{-10} au^2 (same events)	0.01356 for all	130.8 for all	identical
Seed 12, floor 10^{-8}	0.01372	2158	same
Seed 13, floor 10^{-8}	0.01376	213	same

The windows tested are 10^{18} to 2.0×10^{22} g for $q_0 = 10^{-4}$, and none for $q_0 = 1$.

11.2 What the results mean

1. **A typical flyby is far too weak to see.** The median q of 0.014 means the usual wobble is about 1.4 percent of the measurement error. Most flybys are far away, and the Sun's kick cancels most of what they do to the planets.

2. **Direct detection at the** 0.1 m **level is unlikely.** For $q_0 = 1$ there is no mass with at least one expected detection in 26 years. Only about 1 percent of flybys have $q > 1$ at 10^{21} g, so even at the most favorable mass the expected number of detections is well below one. Seeing a PBH flyby this way would need weaker signals to be recoverable.

3. **With weak-signal methods the mass range is open to test.** For $q_0 \leq 10^{-2}$ the window runs from the validity limit, 10^{18} g, up to about 2×10^{22} g. If PBHs made up all of the dark matter in that range, several flybys with detectable signals would be expected in 26 years (the expected number rises as q_0 is lowered). Finding none would count against PBHs of those masses being all of the dark matter, provided the weak signals can really be extracted, which is the assumption behind these thresholds.

4. **The upper edge is set by the flyby rate, not by detectability.** Heavier black holes are rarer, and above about 2×10^{22} g fewer than one flyby through the 50 au disk is expected in 26 years. This is the same mass as `mcMmaxG`, and the same for all small thresholds.

5. **The lower edge is a validity limit, not a result.** The window starts exactly at 10^{18} g because the method is not valid below it (overlapping signals), even though the expected number of detections there is large.

6. **The peak-rate mass is not a robust result.** The tail of q falls roughly as $P(q > x) \propto x^{-1}$ over a wide range (as expected for the tidal-type effect of a distant flyby, once the Sun's kick is subtracted), so the detectable rate Γp is nearly flat over several decades of mass. Where its maximum falls then depends on noise in the extreme tail. The windows are the quantity to report.

7. **The results do not depend on the floor or the random draw.** The floor had no effect because no flyby in the sample comes close enough to a planet for the floor to apply (this would need an impact distance below about 10^{-3} au, which happens in roughly 4×10^{-10} of events). The largest q varies a lot between seeds (131, 213, 591, 2158) because it is set by single close flybys, but the medians and windows do not.

8. **Comparison with Tran et al.** The checks reproduce their numbers: about 1.4 black holes inside 5.2 au at 10^{18} g, and largest values of q of order 10^3 at 10^{21} g. Their conclusion that this method could probe the mass range from 10^{18} g upward is consistent with the windows found here.

11.3 Independent check

An independent Python implementation of the same model, with different random numbers (4×10^5 events), reproduced the notebook: median $q = 0.0138$, largest $q = 497$, no window for $q_0 = 1$, and the windows 10^{18} to 1.58×10^{22} g for $q_0 = 10^{-2}$ and 10^{18} to 2.0×10^{22} g for $q_0 = 10^{-3}$ and 10^{-4}. At $q_0 = 1$ it gave at most about 0.25 expected detections over the whole valid mass range, and at 10^{18} g it gave about 24, 215 and 2100 expected detections for $q_0 = 10^{-2}$, 10^{-3} and 10^{-4}. This confirms that the notebook code implements the model described here. It does not test the model itself against the real Solar System.

12 Limitations

- The thresholds q_0 below 1 assume that signals well below the measurement error can be extracted. Whether that is achievable with real ranging data has not been tested here.
- Part 1 of the notebook uses the prefactor $4\pi\,(\mu/M_\odot)/v_b$ for the impulse. With $GM_\odot = 4\pi^2$ in au and years, the kick $2G\mu/v_b$ used in Part 2 corresponds to $8\pi^2\,(\mu/M_\odot)/v_b$, which is larger by a factor of 2π. Part 2 onward, which produces all the results above, uses the latter.